

Session 6: Loops (For, While, Do-While)Topics:Difference between entry-controlled and exit-controlled loops.Use cases (e.g., menu-driven programs).Example: Countdown timer using loops.

Tasks

1.

Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i;
```

```
    for (i = 10; i >= 1; i--)
```

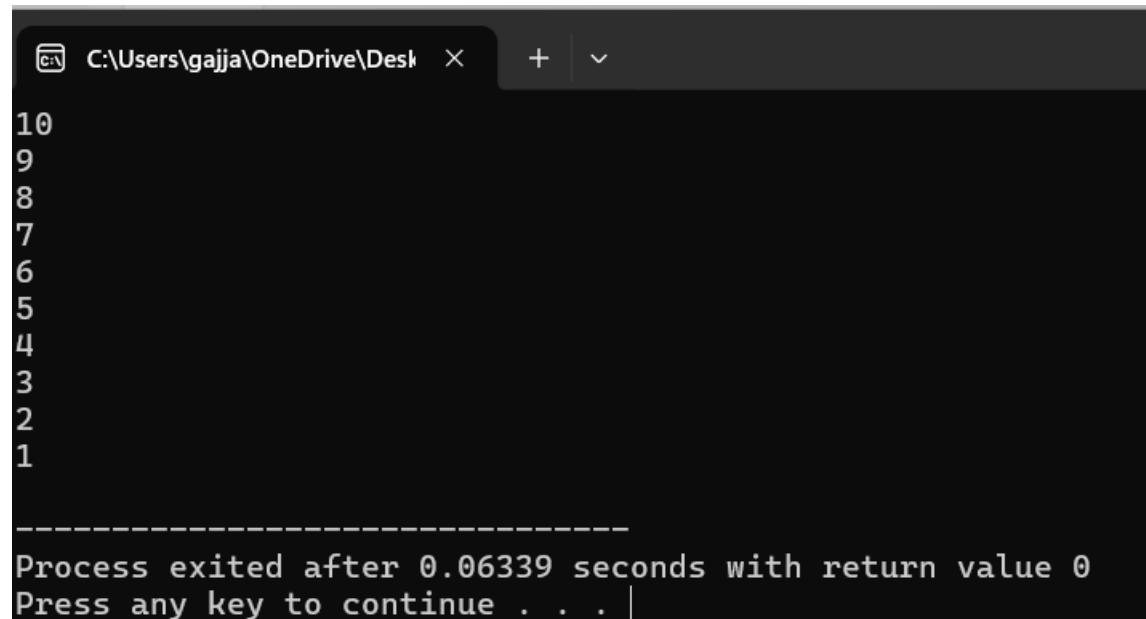
```
    {
```

```
        printf("%d\n", i);
```

```
    }
```

```
    return 0;
```

```
}
```



```
C:\Users\gajja\OneDrive\Desktop >
10
9
8
7
6
5
4
3
2
1
-----
Process exited after 0.06339 seconds with return value 0
Press any key to continue . . .
```

2.

Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit.

Hint: Use input() (or Scanner in Java) to get the user's choice each time.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char teams[4][50] = {
```

```
        "RCB",
```

```
        "CSK",
```

```
        "MI"
```

```
    };
```

```
    char newTeam[50];
```

```
    int choice;
```

```
    int teamCount = 3;
```

```
    int i;
```

```
    int j;
```

```
    while (1)
```

```
    {
```

```
        printf("\n===== IPL MENU =====\n");
```

```
        printf("1. View Favorite 3 IPL Teams\n");
```

```
        printf("2. Add a New Team\n");
```

```
        printf("3. Exit\n");
```

```
printf("Enter your choice: ");  
  
scanf("%d", &choice);  
  
if (choice == 1)  
{  
    printf("\nFavorite IPL Teams:\n");  
  
    for (i = 0; i < teamCount; i++)  
    {  
        printf("%d. %s\n", i + 1, teams[i]);  
    }  
}  
  
else if (choice == 2)  
{  
    if (teamCount < 4)  
    {  
        printf("Enter new team: ");  
        scanf("%s", newTeam);  
  
        for (j = 0; newTeam[j] != '\0'; j++)  
        {  
            teams[teamCount][j] = newTeam[j];  
        }  
  
        teams[teamCount][j] = '\0';  
  
        teamCount++;  
    }  
}
```

```
        printf("Team added successfully!\n");
    }
    else
    {
        printf("Team list is full!\n");
    }
}
else if (choice == 3)
{
    printf("Exiting program...\n");
    break;
}
else
{
    printf("Invalid choice!\n");
}
}

return 0;
}
```

```
C:\Users\gajja\OneDrive\Desk  X + v

===== IPL MENU =====
1. View Favorite 3 IPL Teams
2. Add a New Team
3. Exit
Enter your choice: 1

Favorite IPL Teams:
1. RCB
2. CSK
3. MI

===== IPL MENU =====
1. View Favorite 3 IPL Teams
2. Add a New Team
3. Exit
Enter your choice: 2
Enter new team: GT
Team added successfully!

===== IPL MENU =====
1. View Favorite 3 IPL Teams
2. Add a New Team
3. Exit
Enter your choice: 1

Favorite IPL Teams:
1. RCB
2. CSK
3. MI
```

3.

Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
#include <string.h>
```

```
int main()
{
    char songs[3][50] = {
        "Believer",
        "Perfect",
        "ShapeOfYou"
    };

    char guess[50];
    int randomNumber;
    int correct = 0;

    srand(time(NULL));

    randomNumber = rand() % 3;

    printf("==== Guess the Song =====\n");
    printf("Guess the song name!\n");

    do
    {
        printf("Enter your guess: ");
        scanf("%s", guess);

        if (strcmp(guess, songs[randomNumber]) == 0)
        {
            printf("Correct! You guessed the song!\n");
            correct = 1;
        }
    }
}
```

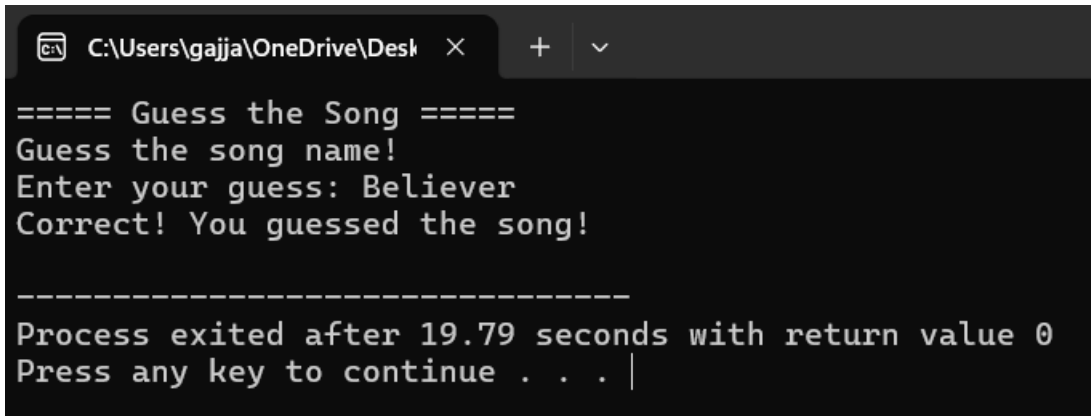
```

    }
    else
    {
        printf("Wrong guess! Try again.\n");
    }

} while (correct == 0);

return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop >
===== Guess the Song =====
Guess the song name!
Enter your guess: Believer
Correct! You guessed the song!

-----
Process exited after 19.79 seconds with return value 0
Press any key to continue . . . |

```

4.

Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

```
#include <stdio.h>
```

```

int main()
{
    int i;

    // Entry-controlled loop using while
    printf("Entry-Controlled Loop (while):\n");

```

```
i = 10;
```

```
while (i < 5)
```

```
{
```

```
    printf("%d\n", i);
```

```
    i++;
```

```
}
```

```
printf("While loop did not execute because the condition was false.\n\n");
```

```
// Exit-controlled loop using do-while
```

```
printf("Exit-Controlled Loop (do-while):\n");
```

```
i = 10;
```

```
do
```

```
{
```

```
    printf("%d\n", i);
```

```
    i++;
```

```
}
```

```
while (i < 5);
```

```
printf("Do-while loop executed once even though the condition was false.\n");
```

```
return 0;
```

```
}
```

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Entry-Controlled Loop (while):

While loop did not execute because the condition was false.

Exit-Controlled Loop (do-while):

10

Do-while loop executed once even though the condition was false.

Process exited after 0.1021 seconds with return value 0

Press any key to continue . . . |